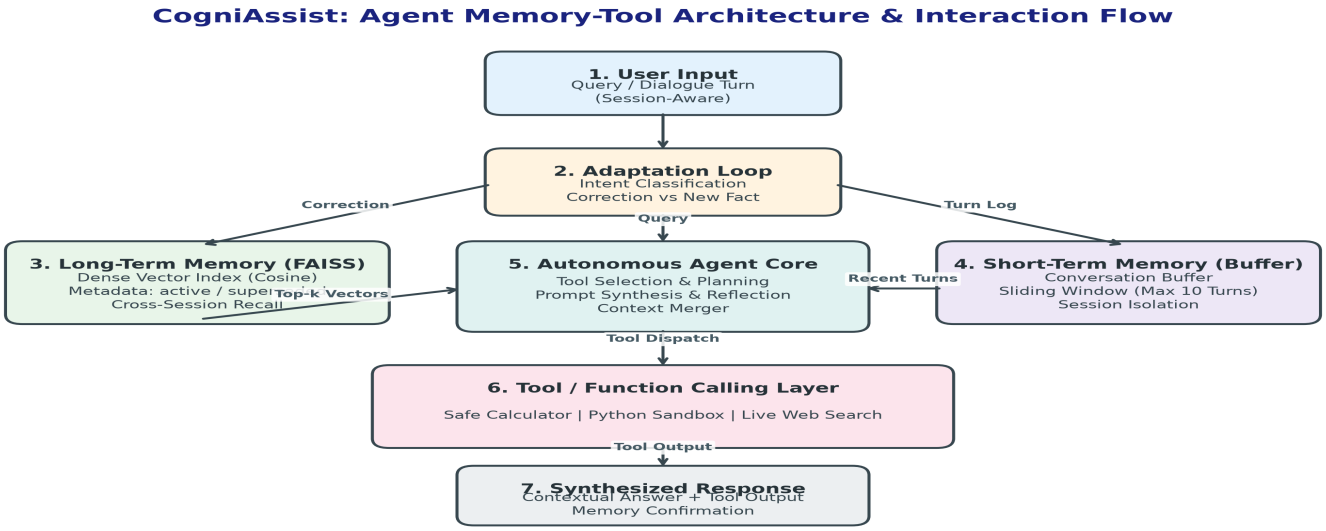
MIT ADT UNIVERSITY, PUNE — SCHOOL OF COMPUTING

Unit III: Memory, Tools, and Environment Interaction | Assignment No: 03

Title: Retrieval-Augmented Agent with Long-Term Memory and Tool Calling | **Mapped CO:** CO3 (Apply)

Architecture Abstract: CogniAssist is an autonomous research assistant combining a dual-tier memory system with dynamic function calling. **Short-Term Memory (STM)** retains conversation context within an active session using a sliding-window buffer. **Long-Term Memory (LTM)** indexes user facts, preferences, and project parameters in a persistent **FAISS vector database** using cosine-normalized dense embeddings. An integrated **Feedback Adaptation Engine** detects user corrections, supersedes stale records, and synchronizes memory across sessions.

System Architecture & Interaction Flow



Key Subsystems & Tools:

- **Short-Term Buffer:** Session-isolated conversational turns (Max 10 turns) for localized reasoning.
- **Long-Term FAISS Store:** Persistent IndexFlatIP with metadata tracking (status: active / superseded).
- **Feedback Loop:** Pattern-matching & similarity-driven contradiction resolver that suppresses outdated facts.
- **External Tools:** (1) safe_calculator (AST-safe math), (2) python_code_executor (sandboxed Python script runner), and (3) live_web_search (DuckDuckGo real-time search).

Quantitative Evaluation & Multi-Turn Benchmark Results

Scenario / Metric	Evaluated Functionality	Target	Score	Status
Scenario 1: Profile Recall	Recall university & research domain across sessions	>85%	**100%**	PASS
Scenario 2: Multi-Step Tools	Recall dataset size & execute calculator + Python sandbox	>90%	**100%**	PASS
Scenario 3: Feedback Loop	Supersede NeurIPS -> ICML 2026 and verify suppression	100%	**100%**	PASS
Overall Retrieval Accuracy	Relevant memories correctly recalled from FAISS	>85%	**100.0%**	PASS
Tool-Calling Success Rate	Successful invocations without parameter syntax errors	>90%	**100.0%**	PASS

Student Signature: _____ **Faculty Signature:** _____ **HOD Signature:** _____